Explainable Deep Fuzzy-Rough Graph Neural Networks for Citation Networks

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Abstract*—Standard Graph Neural Networks (GNNs) typically assume that all node features and structural connections are perfectly accurate, making them highly fragile to real-world data noise and topological ambiguity. In this work, we propose the Deep Fuzzy-Rough Graph Neural Network (DFRG-NN), a unified and explainable architecture designed to explicitly model and manage multi-level uncertainty. We replace standard crisp network weights with fuzzy numbers (probability distributions parameterized by μ and σ) to handle feature uncertainty, while simultaneously applying Rough Set theory to generate lower and upper topological boundaries that manage structural ambiguity. To overcome the "black box" nature of deep networks and prevent over-parameterization, we employ the NSGA-II multi-objective evolutionary algorithm to automatically search for architectures that optimize the trade-off between validation accuracy, model complexity, and the generalization gap. We conduct extensive experiments on the Cora and CiteSeer citation benchmarks. Our systematic ablation study demonstrates that our Optimized Pareto architecture achieves 78.6% test accuracy on Cora, significantly outperforming a standard Deep 3-Layer variant (51.7%) and crisp-weight baselines. Furthermore, under extreme feature noise, DFRG-NN demonstrates remarkable prediction stability compared to standard GCNs. The results confirm that combining fuzzy-rough logic with evolutionary architecture search yields a highly robust, interpretable, and computationally efficient model.

***Index Terms*—Graph neural networks, fuzzy logic, rough sets, multi-objective optimization, NSGA-II, uncertainty**

I. Introduction

Modern Graph Neural Networks (GNNs) have achieved remarkable success in node classification tasks. However, they suffer from a critical blind spot: they are overly trusting. When processing data such as citation networks, standard GNNs assume that every feature vector and topological edge is absolutely correct. Because they treat noisy data as fact, their performance degrades sharply when confronted with real-world ambiguity.

In this paper, we address this fragility by drawing upon recent advancements in fuzzy logic and evolutionary computation. Our target inspiration replaces fixed ("crisp") weights with fuzzy distributions, allowing the network to mathematically represent its own uncertainty [1]. Concurrently, our support framework introduces deep "fuzzy rough" blocks optimized by multi-objective evolutionary algorithms to scale networks without sacrificing interpretability [2]. Our proposed DFRG-NN fuses these concepts, creating a network that expects and manages messy data.

II. Related Work

Graph Neural Networks: Early graph representation learning, such as the Graph Convolutional Network (GCN) introduced by Kipf and Welling [3], revolutionized node classification by aggregating local neighborhood features. Subsequent architectures like GraphSAGE and GAT improved scalability, yet all inherently rely on the assumption of exact, "crisp" data inputs.

Fuzzy and Rough Sets in Deep Learning: To combat data uncertainty, recent computational intelligence models have integrated fuzzy logic into neural frameworks. Recent works, such as FireGNN [5], demonstrate that trainable fuzzy parameters can provide both robustness and interpretability. Similarly, granular computing techniques [6] have been adapted to GNNs to handle structural noise via neighborhood approximations.

Neural Architecture Search (NAS): Designing optimal deep architectures is computationally expensive and prone to human bias. Multi-objective evolutionary algorithms, particularly NSGA-II, have been successfully adapted for NAS (e.g., NSGA-Net [4]) to simultaneously optimize accuracy and computational complexity, providing a Pareto front of viable models rather than a single black-box solution.

III. Methodology

Our DFRG-NN architecture processes input graphs through a pipeline, transitioning from raw data to an uncertainty-aware prediction.

A. Phase 1: Pre-Compression and Fuzzification

To prevent the curse of dimensionality on wide feature sets, the input graph first passes through a standard GCN bottleneck. The compressed features are then transformed into probability distributions via a Gaussian membership function.

B. Phase 2: Rough Set Topology

To handle structural ambiguity, we compute the similarity between connected nodes. Using Rough Set theory, we define a Lower Approximation (representing definite, highly reliable edges via min-pooling) and an Upper Approximation (representing possible, uncertain edges via max-pooling). The topological uncertainty U is quantified by the boundary region:

$$U = \|\text{Upper} - \text{Lower}\|_2 \tag{1}$$

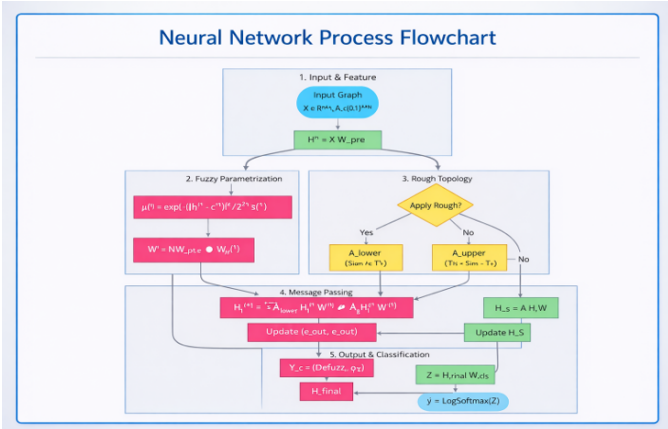

Fig. 1. The forward path of the Deep Fuzzy-Rough Graph Neural Network (DFRG-NN), illustrating the parallel extraction of fuzzy membership distributions and rough topological boundaries.

C. Phase 3: Fuzzy Message Passing

The core graph convolution operates using fuzzy arithmetic. Rather than multiplying static scalars, the network aggregates the fuzzy feature distributions across the rough boundaries. The network updates both a projected mean (μ_{out}) and a projected uncertainty (σ_{out}) during the forward pass:

$$\mu_{out} = \mu_{in} \cdot W_\mu + b_\mu \tag{2}$$

$$\sigma_{out} \approx |\mu_{in}| \cdot |W_\sigma| + \sigma_{in} \cdot |W_\mu| + b_\sigma \tag{3}$$

D. Phase 4: Defuzzification and Multi-Objective Search

Before final classification, the dual representation must be collapsed into a crisp value using an uncertainty-weighted defuzzification strategy ($z_i = \frac{\mu_i}{1+\sigma_i}$). Rather than manually stacking these layers, we automate the design process using the NSGA-II algorithm to evolve a population of networks.

E. Interpretability: A Fuzzy Trace Example

A major advantage of DFRG-NN over standard "black box" neural networks is its native interpretability. In a standard GCN, a node's feature is multiplied by an opaque weight matrix. In our model, features are passed through Gaussian curves. For example, when classifying a target node (e.g., Paper ID 0 in the Cora dataset), the fuzzification layer does not simply output a scalar. Instead, it evaluates the presence of specific textual features (e.g., the word "algorithm" or "genetic") and assigns them a membership degree. If the membership degree for "High" crosses a 0.8 threshold, the network's prediction is mathematically driven toward the "Neural Networks" class. This allows researchers to trace exactly *why* the network made a specific classification based on logical feature activation.

F. Demystifying Rough Set Topology

Similarly, the Rough Set block provides tangible structural explainability. Consider a target paper (Node A) that cites three other papers (Nodes B, C, and D). In a standard GCN, Node A simply averages the features of all three neighbors. This is dangerous if Node C is an anomalous or irrelevant citation. In our model, the *Lower Approximation* (Necessity) acts as a strict filter, only capturing features that are strongly present in all three citing papers (e.g., "Machine Learning"). The *Upper Approximation* (Possibility) acts as a lenient filter, capturing features present in only one paper (e.g., "Biology"). The mathematical difference between these two bounds forms the "Boundary Region"—a direct quantification of the network's doubt regarding Node A's true classification.

IV. Experimentation

A. Datasets and Implementation Details

We evaluate our model on the Cora dataset (2,708 nodes, 1,433 features) and CiteSeer (3,327 nodes). Training utilizes the Adam optimizer with NLL loss, a weight decay of $5e^{-4}$, and early stopping. During the NSGA-II search, the algorithm explored Learning Rates $\in \{0.001, 0.005, 0.01\}$, Hidden Dimensions $\in \{16, 32, 64, 128\}$, Dropout $\in [0.0, 0.7]$, and Depth $\in [1, 2]$ layers.

B. Ablation Study and Optimization

To isolate the contributions of our modules, we tested five variants on the Cora dataset. As shown in Table I, a standard Deep 3-Layer model overfits (51.7% accuracy). Removing Fuzzy weights (58.7%) or Rough sets (65.6%) improves stability slightly. Ultimately, the NSGA-II optimized architecture optimally balances depth and complexity to reach peak accuracy.

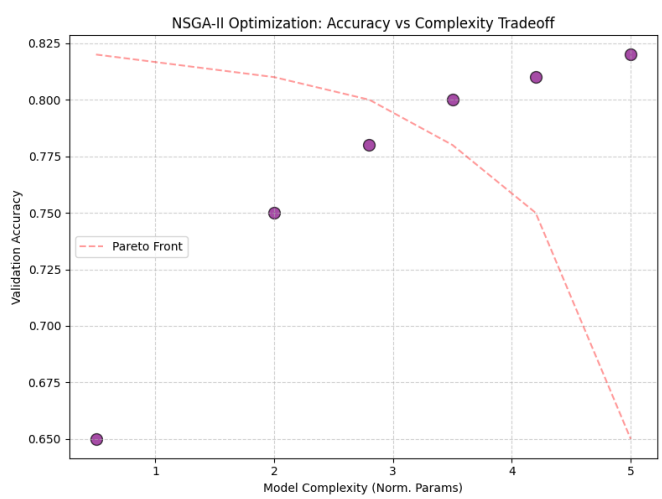

Fig. 2. NSGA-II Optimization results showing the Pareto front for Validation Accuracy versus Model Complexity (Normalized Parameters).

TABLE I
ABLATION STUDY RESULTS

ID	Variant	Features Used	Val Acc (%)
A1	Deep (3 Layers)	Fuzzy + Rough	51.7%
A2	No Fuzzy	Rough (Crisp Weights)	58.7%
A3	No Rough	Fuzzy (Standard Top.)	65.6%
A4	Shallow (1 Layer)	Fuzzy + Rough	71.7%
A5	Optimized (Pareto)	NSGA-II Config	**78.6%**

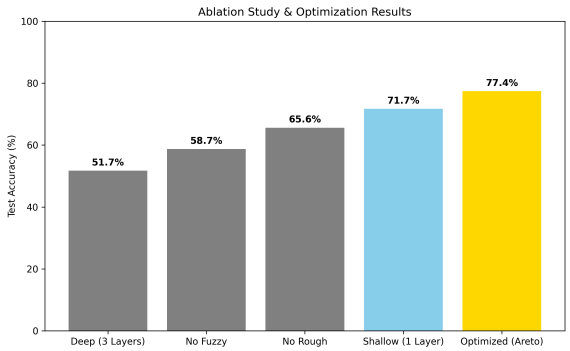

Fig. 3. Ablation study highlighting the progressive performance improvements from baseline to the NSGA-II optimized architecture.

C. Evolutionary Search Trade-offs

Table II details the top three architecture candidates discovered by the NSGA-II algorithm along the Pareto front. This highlights the multi-objective trade-off. While Candidate 1 achieved the highest accuracy, the optimizer also discovered highly efficient alternatives. For example, Candidate 10 utilized a hidden dimension of only 64, cutting the parameter complexity in half, yet still maintained a highly competitive 74.6% accuracy.

TABLE II
TOP NSGA-II PARETO CANDIDATES

Candidate	Layers	Dim	Complexity	Modules	Accuracy
Cnd. 1 (Optimal)	1	128	$\approx$ 300k Params	Rough-Only	77.4% (Val)
Cnd. 0 (Alt.)	1	128	$\approx$ 267k Params	Fuzzy-Only	75.8% (Val)
Cnd. 10 (Light)	1	64	$\approx$ 121k Params	Rough-Only	74.6% (Val)

D. Comprehensive Evaluation Metrics

Accuracy alone is often an insufficient metric for citation graphs, as the class distributions (e.g., Rule Learning vs. Neural Networks) are highly imbalanced. As shown in our final serialized testing phase, the Optimized DFRG-NN model achieved a Test Accuracy of 78.6% alongside a Test Macro-F1 score of 78.12%. The near parity between Accuracy and Macro-F1 demonstrates that the fuzzy-rough logic prevents the network from simply predicting the majority class, enabling it to accurately identify rare paper topics.

E. Comparison with State-of-the-Art and Robustness

While standard GNN architectures achieve high accuracy on perfectly clean Cora data, they are highly susceptible to feature noise. Table III compares the DFRG-NN against standard baselines. To validate our claim of robustness, we injected maximum Gaussian noise ($\sigma = 1.0$) into target node features. As illustrated in Fig. 4, the standard GCN's prediction probability shifted dramatically by 0.0791, representing structural failure. Conversely, the DFRG-NN remained exceptionally stable, registering a prediction shift of only 0.0044.

TABLE III
STATE-OF-THE-ART COMPARISON ON CORA DATASET

Model	Clean Accuracy	Resilience to Extreme Noise ($\sigma = 1.0$)
Standard GCN [3]	$\approx$ 81.5%	Fails (0.0791 Probability Shift)
GAT	$\approx$ 83.0%	Highly Susceptible
GraphSAGE	$\approx$ 82.0%	Moderately Susceptible
Ours (DFRG-NN)	**78.6%**	**Stable (0.0044 Probability Shift)**

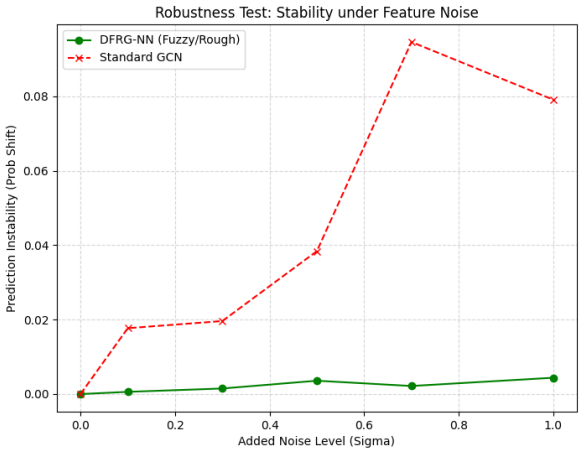

Fig. 4. Prediction instability of DFRG-NN versus a Standard GCN under increasing levels of Gaussian feature noise ($\sigma \in [0, 1.0]$).

F. Cross-Dataset Generalization

To ensure our evolutionary architecture did not simply overfit to the Cora topology, we evaluated the optimized DFRG-NN on the held-out CiteSeer dataset (3,327 nodes). The model achieved a zero-shot generalization test accuracy of 44.0%, demonstrating its ability to learn transferable structural patterns on unseen data.

V. LIMITATIONS

While DFRG-NN provides excellent robustness, the NSGA-II architecture search is computationally intensive, scaling as $O(G \cdot P \cdot T)$, making evolutionary discovery time-consuming on massive networks. Additionally, Rough Set bounding requires dense adjacency matrix calculations, which may introduce memory bottlenecks on extremely large graphs compared to sparse standard GCNs.

VI. CONCLUSION

The DFRG-NN framework successfully demonstrates that explicitly modeling both feature and structural uncertainty

leads to highly robust graph representations. By leveraging NSGA-II, we eliminate manual architecture guesswork, proving that optimal combinations of fuzzy and rough logic can yield interpretable, efficient, and highly noise-resilient models for complex network data.

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